Observational Studies with Covariates

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For previous notes, see
https://oftrongs.github.io/Causal/notes/Causal_Two_points_of_view.pdf

In previous discussion

But we did not mention the information of covariates...

Role of Covariates

Many contents comes with the information of covariates:

- power to estimate treatment effect under non-random-treatment-assignment situation
- conditional-on-covariates assumptions(unconfoundedness and overlap)
- possibility to estimate heterogeneous effect CATE, [?]and to estimate ATE by CATE
- propensity score
- covariate balance (related with overlap)

how we understand the i.i.d sample

- In observational point of view, we view the observed units as samples from an unknown **super population**, which we make assumptions and inference about.
- This is like the traditional statistical analysis, with sample drawn IID from the population.

$$\{X_i, Z_i, Y_i(1), Y_i(0)\}_{i=1}^n \overset{\text{i.i.d.}}{\sim} \{X, Z, Y(1), Y(0)\}.$$

  But we can not observe its sample realization fully due to the **fundamental missing problem**.
- What we can observe sample from?

$$Y = ZY(1) + (1 - Z)Y(0),$$
$$\{X_i, Z_i, Y_i\}_{i=1}^n \overset{\text{i.i.d.}}{\sim} \{X, Z, ZY(1) + (1 - Z)Y(0)\}.$$

  where Y is a nonparametric function of the original population of $\{Z, Y(1), Y(0)\}$.

how we understand the i.i.d sample

For this super-population framework, we can always adopt a **data-generating point of view**.

There are many ways to understand the sample data $\{X_i, Z_i, Y_i\}_{i=1}^n$, as we can factorize the joint distribution $\Pr(X, Z, Y(1), Y(0))$ in many ways:

- $\Pr(X, Z, Y(1), Y(0))$
- $\Pr(X) \times \Pr(Z, Y(1), Y(0) \mid X)$
- $\Pr(X) \times \Pr(Y(1), Y(0) \mid X) \times \Pr(Z \mid X, Y(1), Y(0))$
- $\Pr(X) \times \Pr(Z \mid X) \times \Pr(Y(1), Y(0) \mid X, Z)$

Assumptions[1] on super population

- Unconfoundedness

$$Y(z) \perp\!\!\!\perp Z \mid X \quad \text{for} \quad z = 0, 1$$

- Overlap

$$0 < P(Z = 1 \mid X) < 1$$

[1] there are other version of the assumptions

Role of Overlap

- see `https://oftrongs.github.io/Causal/notes/Overlap_and_Balance.pdf`

Role of Unconfoundedness

Under unconfoundedness assumption, we

- reach 'random assignment' within subpopulations defined by values of observed covariates $X = x$

Also, we have

$$Pr(Y \mid X, Z = z) = Pr(Y(z) \mid X) \quad \text{for } z = 0, 1$$

which implies

- the observable distribution of Y in each treatment arm $Z = z$ equals the distribution of the potential outcome $Y(z)$, given covariates X
- two forms of identification with X
  - $E(Y(1) \mid X) = E(Y \mid X, Z = 1)$, $E(Y(0) \mid X) = E(Y \mid X, Z = 0)$
  - $E(Y(1) \mid X) = E(\frac{ZY}{e(X)})$, $E(Y(0) \mid X) = E(\frac{(1-Z)Y}{1-e(X)})$

Role of Unconfoundedness

Recall the ways of decomposition of the joint distribution

$$\begin{aligned}\Pr(Z, Y(1), Y(0) \mid X) &= \Pr(Y(1), Y(0) \mid X) \times \Pr(Z \mid \cancel{Y(1), Y(0)}, X) \\ &= \Pr(Z \mid X) \times \Pr(Y(1), Y(0) \mid \cancel{Z}, X)\end{aligned}$$

So we actually reduce the question to two separate conditional distributions: (1) **the outcome model** and (2) **the propensity model**.

$$\Pr(Z, Y(1), Y(0) \mid X) = \Pr(Z \mid X) \times \Pr(Y(1), Y(0) \mid X)$$

Then, what we do in causal inference:

- make assumptions on $\Pr(Z \mid X)$ and $\Pr(Y(1), Y(0) \mid X)$, and
- infer the characteristics of $\Pr(Y(1), Y(0) \mid X)$ (e.g. ATE and CATE)

Two estimands: ATE and CATE

- Average treatment effect(ATE)

$$\tau = E(Y(1) - Y(0)) = E(Y(1)) - E(Y(0))$$

  - important when we only care about the average causal effect for the target population (i.e. care more about $\{Y(1), Y(0)\}$ instead of $\{Y(1), Y(0) \mid X\}$)
  - a scalar value, so (1) easy to define bias and variance; and (2) often do not necessarily need a model assumption
- Conditional average treatment effect(CATE)

$$\tau(X) = E(Y(1) - Y(0) \mid X) = E(Y(1) \mid X) - E(Y(0) \mid X)$$

  - important when we care about average causal effect conditional on the value of X for the target population(i.e. we care about $\{Y(1), Y(0) \mid X\}$)
  - a function-type estimand, so (1) not straightforward to define the properties of estimators(point-wisely or population-wisely); and (2) [?]usually need to make a model assumption to better define and analyze the properties of an estimator

Homogeneous or Heterogeneous?

Based on the scientific nature of our problem, we can assume:

- homogeneity
- heterogeneity

Response Surfaces

- Let $\{X, Y(1), Y(0)\} \mid Z = z \sim P_z$, in each P_z, the conditional expectation of the variable Y given a particular value of X is called the response surface. (Rubin 1973)

$$R_z(x) = E(Y \mid X = x, Z = z) = E(Y(z) \mid X = x, Z = z)$$

- Under unconfoundedness,

$$R_z(x) = E(Y(z) \mid X = x, Z = z) = E(Y(z) \mid X = x) = \mu_z(x)$$

- The difference in response surfaces at $X = x$, is the effect of the treatment variable at $X = x$.

$$\tau(x) = R_1(x) - R_0(x) = E(Y(1) - Y(0) \mid X = x)$$

- "parallel response surfaces" means homogeneous effect

Estimators

In the following parts, we talk about some estimators[2]:

- Propensity-Weighted estimator(e.g. IPW) for ATE
- Regression-based estimator for ATE and CATE
- Regressed-residuals-form estimator (PLM and heterogeneous PLM) for ATE and CATE
- DR estimator
- Meta-learners for CATE

When the model assumes homogeneous treatment effect, the estimand is ATE. Otherwise we talk about CATE.

[2]Note that the estimator itself is just a calculation of sample data, and does not necessarily indicate model assumptions, but we need to analyze its properties with specific better-scientific-plausible model assumption.

IPW - a nonparametric estimator for ATE

- Recall the identification form of ATE

$$\tau = E\left(\frac{ZY}{e(X)} - \frac{(1-Z)Y}{1-e(X)}\right)$$

which motivates the following nonparametric(moment-based) estimator[3],

$$\hat{\tau}_{ipw} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{Z_i Y_i}{\hat{e}(X_i)} - \frac{(1-Z_i)Y_i}{1-\hat{e}(X_i)}\right)$$

- $\hat{\tau}_{ipw}$ is consistent for τ if $\hat{e}(x)$ is consistent for $e(x)$ (with strict overlap[..])

[3] An oracle version with true propensity score is $\hat{\tau}^*_{ipw} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{Z_i Y_i}{e(X_i)} - \frac{(1-Z_i)Y_i}{1-e(X_i)}\right)$, and $\sqrt{n}(\hat{\tau}^*_{ipw} - \tau) \sim N(0, V^*_{IPW})$(Wager 2024)

Outcome Regression - a parametric estimator for ATE with model assumption

- Recall the identification form of $E(Y(z) \mid X) = E(Y \mid X, Z = z)$, let $\mu_z(X) = E(Y(z) \mid X)$, $z = 0, 1$
- If we specify a model with parameter α, then we can estimate by $\hat{\mu}_z(X) = \mu_z(X; \hat{\alpha})$.
- Outcome modeling
- OR-estimator of ATE is

$$\hat{\tau}_{reg} = \frac{1}{n} \sum_{i=1}^{n} (\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i))$$

or

$$\hat{\tau}_{reg} = \frac{1}{n} \sum_{i=1}^{n} \left\{ Z_i \big(Y_i - \hat{\mu}_0(X_i) \big) + (1 - Z_i) \big(\hat{\mu}_1(X_i) - Y_i \big) \right\}$$

Outcome Regression - Linear Model without Heterogeneity

$$\mu_z(X) = \beta_0 + \beta_Z z + \beta_X^\top X, \quad z \in \{0, 1\}$$
$$\tau = \tau(X) = \beta_Z$$

- pooled-estimator $\hat{\tau}$ coefficient of Z by regression $Y \sim Z + X$, which is an OLS estimator so unbiased and consistent(assumption of regression?)(ANOVA)
- equivalent calculation forms()
- ? pooled-estimator same as non-pooled diff
- why we use X here?

Outcome Regression - Linear Model with Heterogeneity

$$\mu_z(X) = \beta_0 + \beta_Z z + \beta_X^\top X + \beta_{ZX}^\top X z, \quad z \in \{0, 1\}$$

$$\tau(X) = \beta_Z + \beta_{ZX}^\top X$$

$$\tau = \beta_Z + \beta_{ZX}^\top E(X)$$

- CATE: regress $Y \sim Z + X + ZX$, and $\hat{\tau}(X) = \hat{\beta}_Z + \hat{\beta}_{ZX}^\top X$
-

Outcome Regression - General Form of Model Assumption

Can we allow arbitrary form?

DR

Partial Linear Model

Partial linear model allows Y to have arbitrary dependence on X and linear with Z. (Robinson 1988)

$$E(Y \mid X, Z) = \beta Z + \theta(X)$$

i.e.

$$Y - E(Y|X) = \beta(Z - E(Z|X)) + U,\ E(U|X,Z) = 0$$

In causal settings, given unconfoundedness, $\tau = \beta$

- regression-of-residuals estimator $\hat{\tau} = S^{-1}_{Z-\hat{Z}} S_{Z-\hat{Z},Y-\hat{Y}}$
- double-kernel estimator is consistent $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 \Phi^{-1})$. if

R-learner

Summary

for more notes, see `https://oftrongs.github.io/Causal.html`

references

Robinson, Peter M (1988). "Root-N-consistent semiparametric regression". In: *Econometrica: journal of the Econometric Society*, pp. 931–954.

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